

Assembling Two Parts in One Hand

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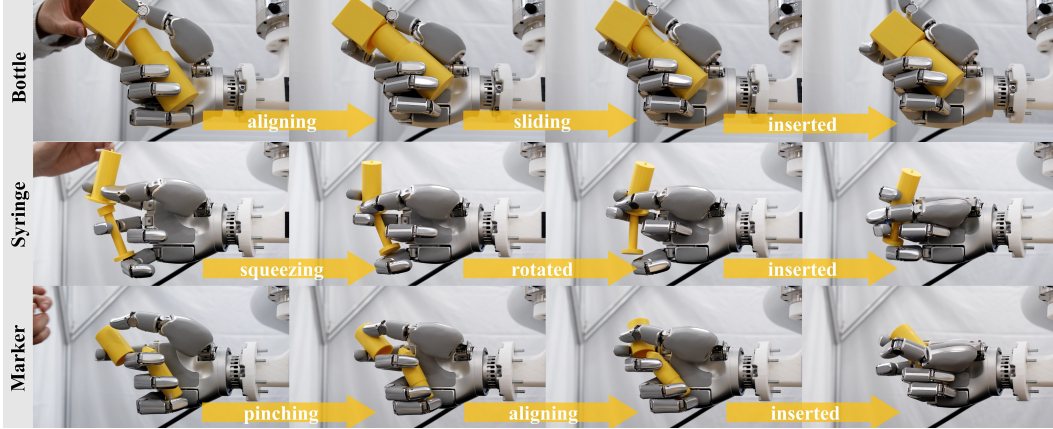


Figure 1: **Real-world in-hand assembly.** A single dexterous hand mates two objects using fine finger coordination without a second arm or external fixture, shown here on the Bottle, Syringe, and Marker tasks.

Abstract: A hallmark of human dexterity is the cooperative use of fingers, where different fingers take on distinct yet coordinated roles to accomplish fine manipulation, such as capping a pen with the hand that holds it. We study this finger-level coordination through *in-hand assembly*: mating two rigid objects within a single dexterous hand, with no second arm and no fixture. We present a reinforcement learning formulation to solve this problem in a unified framework, which is driven by a goal relative pose between the two parts. Finger coordination is shaped by a function-based auxiliary reward and regularized toward a single human reference pose, while domain randomization and a fusion of historical proprioception and object observation confer robustness to occlusion-induced estimation noise. The same recipe solves three different assembly tasks (Bottle, Syringe, and Marker). Trained purely in simulation, the policies transfer zero-shot to hardware with a single camera, demonstrating robustness to state-estimation errors caused by occlusion. Our experiments also reveal that in-hand assembly places demands on hand morphology, as a benchmark for modern robotic hand systems.

Keywords: Dexterous Manipulation, In-Hand Assembly, Reinforcement Learning, Sim-to-Real

1 Introduction

While animals can also perform basic manipulation behaviors with their limbs or digits, such as pushing or pulling objects, the ability to execute complex manipulation through coordinated finger interactions is most refined in human hands [1, 2]. For instance, humans squeeze a syringe with one hand, cap a pen with the hand that holds it, or press a lid back onto a bottle held in the same hand. This cooperative use of individual fingers is common in everyday human manipulation, whereas current work on dexterous manipulation remains limited to relatively simple or repetitive finger motions and contact modes, as exemplified by pick-and-place tasks [3, 4, 5, 6] and in-hand reorientation [7, 8, 9]. These capabilities have yet to fully exploit the dexterity of modern robotic hands.

To move beyond these limited settings, we study tasks that more directly reflect and exploit the coordinative dexterity of human hands. Specifically, we focus on *in-hand assembly*: tasks in which a single multi-fingered hand simultaneously manipulates two objects to mate them and operates the assembly under articulation constraints (Fig. 1). In-hand assembly tasks, however, are inherently challenging in several aspects. First, the tasks are contact-rich and control-intensive, requiring forceful yet precise coordination across four or five fingers. Such intricate multi-finger motions are difficult to obtain through teleoperation [10] or finger-by-finger kinesthetic demonstration [11]. Second, assembly tasks exhibit diversity in object geometry, mating mechanisms, and target motions. Together with the high degree of freedom in the control space, this often leads to task-specific formulations and optimization objectives, posing a major scalability challenge. Third, assembly requires precise alignment, while occlusion-induced errors in object state estimation pose significant challenges to control robustness.

To address these challenges, we develop a policy learning approach based on reinforcement learning (RL) in simulation to automatically explore solutions. From the perspective of relative object state goal reaching, we abstract a shared task formulation that captures reusable structure across assembly tasks, with different roles assigned to different fingers, such as object translocation and compliant support, triggered by a function-relevant reward. We further propose reference snapshots from human manipulation to constrain exploration. To enable robust closed-loop execution under occlusion, we apply observation domain randomization during training, encouraging the policy to implicitly calibrate noisy estimates through proprioceptive feedback.

Our contributions are threefold: (1) **In-hand assembly as a dexterity benchmark**. We introduce in-hand assembly as a challenging dexterous manipulation setting. By requiring one hand to coordinate multiple fingers across two interacting objects, it provides a task family for evaluating coordinative dexterity in robotic manipulation systems. (2) **Systematic solution**. We present a systematic solution for in-hand assembly that integrates simulation-based RL, finger function rewards, and human kinematic priors, which is unified across tasks. These components make up the first in-hand assembly demonstration. (3) **Real-world effectiveness**. The system performs real-world in-hand assembly using only stereo vision from a single camera and historical proprioception, demonstrating effectiveness under state-estimation noise and unexpected disturbance.

2 Related Work

2.1 Dexterous Manipulation with Multi-Fingered Hands

Most work on multi-fingered hands centers on grasping, either synthesizing large-scale datasets [3, 12, 13] or training policies to grasp novel objects [14, 15, 16, 17]. Such grasps often reduce the hand to enveloping a single object with all fingers, which a low-DoF gripper can also achieve [18]. A smaller body of work grasps several objects with one hand [19, 20, 21], but partitions the fingers into subsets that each enclose a separate object, acting as independent grippers that co-hold the objects rather than coordinating them, leaving the coordinative dexterity limited. Another line uses sim-to-real RL [22, 23] for in-hand manipulation of a single object, typically on easy-to-formulate skills such as in-hand rotation [7, 8, 9, 24, 25, 26] or translation [27]. Others learn from teleoperated demonstrations [11, 28, 29, 30] or human motion retargeting [31, 32], but the embodiment gap between human and robot hands hinders agile, highly dexterous skills with high precision. Overall, existing systems still do not capture the human-like dexterity of precisely coordinating multiple objects within a single hand. In contrast, we use multi-object in-hand assembly to expose this gap and probe the boundary of machine dexterity with modern multi-fingered hands.

2.2 Robotic Assembly

Assembly is a long-standing problem in robotic manipulation, with insertion (peg-in-hole) as its canonical instance. Current learning-based methods show that sim-to-real RL can acquire robust contact-rich insertion policies [33, 34, 35, 36]. Almost all of this work, however, uses a parallel-jaw gripper on a robot arm: the gripper rigidly clamps one part while the other is held by a fixture or a second arm. Lee et al. [37] perform keyhole peg-in-hole with a dual-arm system equipped with robotic hands in which the fingers only grasp and pre-rotate each part. More broadly, two-object

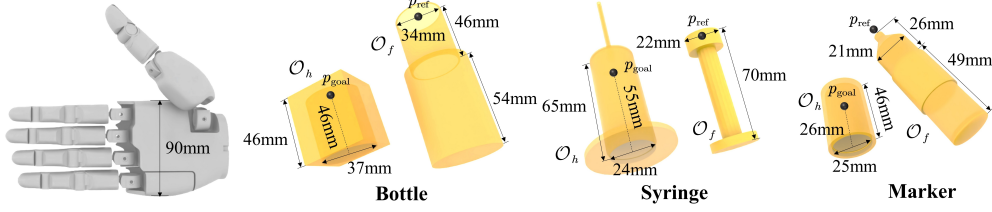


Figure 2: **Task illustration with key geometric parameters.** This figure includes the semi-transparent parts of three task instances, key geometric parameters, and the positions of the reference point p_{ref} and goal point p_{goal} to define a goal-reaching reward (see §3.4).

contact-rich skills with multi-fingered hands are performed bimanually, e.g., twisting lids off bottles with one hand stabilizing and the other turning [38], and target disassembly rather than the harder mating of parts. We take inspiration from these canonical assembly tasks but move the entire mating process into a single dexterous hand, thereby providing a direct benchmark for the intrinsic dexterity of multi-fingered hands.

3 Method

We formulate *in-hand assembly* as a reinforcement learning problem (§3.1), and introduce three tasks for this setting (§3.2). Based on this, we design our unified observation & action spaces (§3.3), and reward structure (§3.4). We also involve human manipulation snapshots for initialization (§3.5), and apply domain randomization during training to improve robustness (Appx. A). The simulation setup especially for contact-rich assembly [33], RL implementation details, and policy architectures are postponed to Appx. A.

3.1 Problem Formulation

Let $\mathcal{O}_h$ and $\mathcal{O}_f$ be the two rigid bodies to be mated, and let $\mathcal{H}$ be a *single* dexterous hand with n_q revolute joints. At every control step, the hand must drive $\mathcal{O}_h$ into a *goal relative pose* with respect to $\mathcal{O}_f$ that satisfies a task-specific geometric mating constraint. The names (held versus fixed) follow the assembly literature [34], whereas no second arm, table fixture, or vise is available; both objects are supported by the same hand.

We partition $\mathcal{H}$'s fingers into two role sets and regularize their motion using separate reward components (see §3.4) that softly constrain which fingers contact which object. The resulting behavioral specialization (reported in §4) is: the held set $\mathcal{F}_h = \{\text{thumb, index}\}$ manipulates $\mathcal{O}_h$ at its top and provides the fine alignment and goal-pose-reaching motion. The fixed set $\mathcal{F}_f = \{\text{middle, ring, little}\}$ cages $\mathcal{O}_f$ and resists the reaction wrenches generated by $\mathcal{F}_h$, while also assisting with the alignment and assembly, e.g., the middle and ring fingers might fine-tune the pose of $\mathcal{O}_f$, and the pinky finger can assist with insertion.

Casting this as a partially observable Markov decision process (POMDP) $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \gamma)$, we seek a recurrent policy $\pi_\theta(a_t \mid o_{\leq t})$ that maximizes $\mathbb{E}_\pi \left[\sum_{t=0}^{T_{\max}} \gamma^t R_t \right]$ subject to joint-limit and contact-feasibility constraints.

3.2 Task Instances

We have instantiated in-hand assembly into three tasks: (1) Bottle cap and bottle assembly (Bottle); (2) Syringe insertion (Syringe); (3) Marker cap fitting (Marker), as shown in Fig. 2. The primary differences among these three assembly tasks lie in the parts' geometries, assembly hand pose, and tolerances. For instance, the parts in the Bottle task are relatively large, which can lead to unstable grasping, and adjusting the bottle's orientation (often necessary) requires the involvement of each finger's motion. In the Syringe task, the plunger requires contact with the back of the ring finger and the involvement of the pinky finger for insertion, while, in the Marker task, the hand causes extensive occlusion, placing demands on the robustness of the state estimator and the policy against estimation errors. The assembly types for all three tasks are similar to those described in previous assembly literature [34, 37], specifically plug-in assembly. In the future, we plan to expand to other

types of assembly, such as threading and friction-fit assemblies; however, these will require more accurate simulation [39].

3.3 Observation and Action Spaces

Observation. The policy reads a 34-dimensional vector

$$o_t = [p_t^{(h)} | \hat{z}_t^{(h)} | p_t^{(f)} | \hat{z}_t^{(f)} | q_t] \in \mathbb{R}^{3+3+3+3+22}, \quad (1)$$

where $p^{(\cdot)} \in \mathbb{R}^3$ is the object centroid in the hand-base frame, $\hat{z}^{(\cdot)} \in S^2$ is the unit vector representing the object’s body-frame z -axis, which is the object’s axis of rotational symmetry, in the world coordinates, $q_t \in \mathbb{R}^{22}$ are the measured joint angles. If we randomize hand tilt during training, the observation space is extended to include ϕ, θ , the roll and pitch pose of the hand in the world frame.

We note that the observation space consists solely of the incomplete state of the object derived from visual state estimation and the robot’s proprioception; it does not include any higher-order quantities such as joint velocities or contact force signals. Since most of the components being assembled are rotationally symmetric about their body z -axis, we omit the estimation of this degree of freedom. The policy is represented as a recurrent neural network (RNN) to tackle the partial observability.

Action. The action $a_t \in [-1, 1]^{22}$ is a normalized joint-position-target delta applied to a stiff implicit Proportional-Derivative (PD) controller:

$$q_{t+1}^{\text{target}} = \text{clip}(q_t + \alpha a_t, q_{\min}, q_{\max}), \quad \alpha = 0.1 \text{ rad}, \quad (2)$$

where q_t is the current PD joint-position target, and $q_{\min}/q_{\max}$ represents the lower/upper joint limit. We control at 15 Hz, and this action structure results in naturally smooth absolute joint angles.

Real-World Object Pose Tracking. At deployment, we recover the 6-DoF poses of $\mathcal{O}_h$ and $\mathcal{O}_f$ from an RGB-D stream with FoundationPose [40], which registers each object mesh against the first frame and refines the pose per frame thereafter. Each pose is then reduced to the $(p, \hat{z})$ form of Eq. (1); a lightweight depth-consistency gate rejects per-frame estimates whose predicted depth disagrees with the measured depth at the projected center, which prevents the tracker from locking onto the manipulating hand when it occludes the object. More details can be found in Appx. B.

3.4 Goal-Reaching with Auxiliary Rewards

Goal-reaching reward. The main reward signals come from:

$$R_t^{\text{goal}} = 5 \cdot \underbrace{(\exp(-60e_{xy}) - 60e_z - 5e_\theta)}_{\in[0,1]} + \underbrace{\exp(-200e_{xy} - 120e_z - 30e_\theta)}_{\in[0,1]}, \quad (3)$$

where e_{xy} and e_z are the xy -plane and z -axis distance between the reference point and the goal point illustrated by Fig. 2 in $\mathcal{O}_f$ ’s local frame (in meters), and $e_\theta = 1 - z_{\text{held}} \cdot z_{\text{fixed}}$, with z_{held} and z_{fixed} being the normalized direction vector of body- z axis of $\mathcal{O}_h$ and $\mathcal{O}_f$, respectively. This reward describes the achievement of relative spatial relationship between the two parts, which is the objective of the assembly task. We set up two additive terms to provide goal-reaching motivations at different scales.

Auxiliary reward. Active throughout the episode, it is a multiplicative combination of three normalized quality signals:

$$R_t^{\text{aux}} = \underbrace{\rho_t^{\text{pinch}}}_{\in[0,1]} \cdot \underbrace{\rho_t^{\text{grasp}}}_{\in[0,1]} \cdot \underbrace{\rho_t^{\text{pose}}}_{\in[0,1]}. \quad (4)$$

ρ_t^{pinch} counts how many of {thumb, index} are in contact with $\mathcal{O}_h$; $\rho_t^{\text{grasp}} = \exp(-20 \cdot \max_{i \in \mathcal{F}_f} d_i)$ rewards $\mathcal{F}_f$ caging $\mathcal{O}_f$; $\rho_t^{\text{pose}} = \exp(-\|q_t - q_0\|_M^2/12)$ constrains drift from a nominal grasp pose q_0 under a per-joint mask M . These rewards reflect an empirical expectation finger organization: the thumb and index finger work together to manipulate the held object (so-called a “pinch”), while the middle, ring, and pinky fingers manipulate the fixed object (a “grasp”). The pose constraint is inspired by prior work that learns from human video, where extracted references guide wrist

156 motion and grasping pose [41, 42]. For in-hand assembly, however, we apply such references only
 157 as single-frame snapshots: the fine, intricate finger motion of in-hand manipulation is susceptible to
 158 the embodiment gap and extraction noise, so multi-frame references offer little usable guidance for
 159 finger motion. The reward scheme leaves the fine-grained finger motion to the goal-reaching reward.

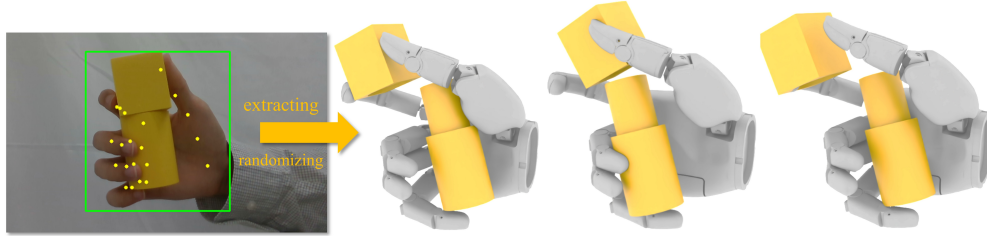


Figure 3: **State initialization from a human reference snapshot.** On the left is the human reference snapshot for the Bottle task; the three simulated states on the right are initial state examples generated based on the reference hand pose. Initialization for other tasks is postponed to Fig. ??

160 3.5 Snapshot as Initialization

161 Since the hand is anchored and cannot grasp from a tabletop, every episode must start with the
 162 objects already in hand. From a single human-demonstration snapshot of the target task, we geo-
 163 metrically retarget [10] to generate a robotic hand state based on the human hand pose with a 30%
 164 reduction of joint angles of the index finger to widen the opening formed by the index finger and
 165 thumb. We then apply the state estimator to extract the objects’ poses. The two sources are inde-
 166 pendent, so directly composing them usually produces hand-object penetration. We resolve this and
 167 inject randomization in initial states with a settling phase: holding the hand as a fixed collider, we
 168 apply a small random perturbation force and torque to each object and step the simulator for a few
 169 steps; the perturbation is then removed, and the episode begins from the resulting state. This proce-
 170 dure settles the objects into randomized configurations, crucially broadening the state coverage seen
 171 during training. A more detailed randomization procedure is postponed to Appx. A. The objects’
 172 gravity is removed for the first few steps in an episode to prevent them from quickly falling. In this
 173 way, the policies learn to establish contacts with the objects in these warm-up steps.

174 4 Results

175 4.1 Experimental Setup

176 We use one 22 Degree-of-Freedom (DoF) Sharpa Wave Hand [43] in most of the simulation and real-
 177 world experiments, which is fixed on a Franka Research 3 arm [44]. The simulation experiments
 178 are conducted in IsaacSim [45]. We rely on a single RealSense D435 camera [46] to perform state
 179 estimation. We send control commands to the robot via a Linux workstation with an RTX-3090
 180 Graphics Processing Unit (GPU). See Appx. B for more details of real-world deployment.

181 4.2 In-Hand Assembly as a Hand Morphology Benchmark

182 Intuitively, in-hand assembly places stronger demands on the morphology of the robotic hand than
 183 prior in-hand manipulation tasks, which can be accomplished by a wider variety of hands [9]. We
 184 test this hypothesis by transferring the same pipeline to different hands in simulation. Beyond the
 185 Sharpa Hand used in our main experiments, we evaluate the Wuji Hand [47], the Allegro Hand [48],
 186 and the XHand [49]. Fig. 4 reports the distribution of reference-point errors on the Syringe task.
 187 These errors are obtained from trials that do not early terminate.

188 From the results, we can tell that the Sharpa Hand and Wuji Hand perform best, owing to their
 189 human-like dimensions, high DoF, and wide joint ranges. The Allegro Hand has only four fingers
 190 and is larger than a typical human hand, but each finger has high degrees of freedom and a wide
 191 joint range, enabling it to align objects in the xy-plane; the lack of a fifth finger, however, prevents
 192 the deep insertion required along the z-axis. The XHand is close to human-hand size but has fewer
 193 degrees of freedom—most of its fingers lack abduction/adduction joints—and it fails to insert the
 194 objects, and the objects cannot be aligned in the xy-plane in many cases.

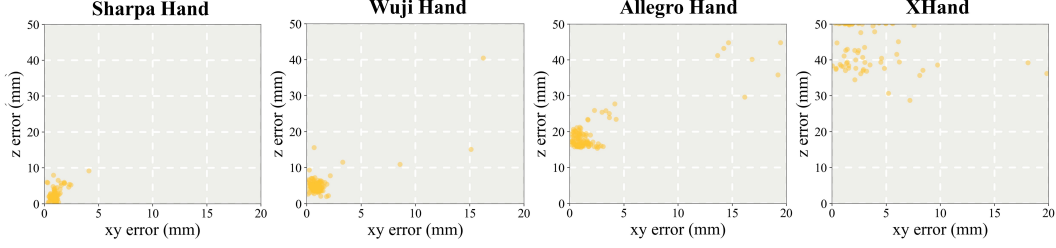


Figure 4: **Goal-reaching errors for different hand morphology.** The yellow points in the figures are error samples with a total number of 128.

195 4.3 Simulation Ablation

196 We compare our policy against the following variants of our method: 1) A proprioception-only
 197 policy. This is a recurrent policy that does not incorporate object state estimation. 2) A historical
 198 input-output MLP policy (history MLP) [50]: instead of using an RNN, this policy uses 8 frames
 199 of historical policy input-output data as policy observations, implemented as an MLP. 3) A policy
 200 that does not use the finger-function-based contact reward $\rho^{\text{pinch}} \cdot \rho^{\text{grasp}}$ during training (w/o finger
 201 reward). 4) A policy that does not use the reference-pose-based reward ρ^{pose} during training (w/o
 202 pose reward). The results in Fig. 5 show the performance metric calculated using the cumulative
 203 goal-reaching reward across episodes of the models obtained at epoch 600 across 3 independent
 204 training runs. Episodes that end within 0.5s of the warmup are not included in the calculation, as
 205 they are considered to start in a poor or infeasible state.

206 The results show that differ-
 207 ent tasks demonstrate different
 208 requirements for vision, and
 209 removing vision generally re-
 210 duces task performance. For
 211 instance, removing vision sig-
 212 nificantly impacts the perfor-
 213 mance of the Bottle task. Prior
 214 work indicates that certain in-
 215 hand manipulation skills can be
 216 performed well with propriocep-
 217 tion alone [8, 9]. Assembly,
 218 however, sometimes requires the
 219 policy to have ready access to
 220 the spatial relationship between
 221 the two objects, so having this
 222 information makes learning eas-
 223 ier and makes it feasible to ma-
 224 nipulate the two objects in pre-
 225 cise relative alignment based on
 226 explicit object states. We also find that LSTM achieves better performance than stacking historical
 227 information within a fixed sample budget, implying that recurrent policies yield a better solution
 228 to POMDP. Meanwhile, we observe that removing some components of the reward affects learning
 229 efficiency and the policies’ rollout behavior at convergence. This is because the system has a high
 230 degree of operational freedom, so that many behaviors can yield goal-reaching rewards, but some of
 231 them are unnatural, do not perform well for all initial object states due to suboptimal contact modes,
 232 and thus being shallower local minima of policy optimization.

233 4.4 Quantitative Evaluation with Estimation Error

234 We evaluate the goal-reaching reward (as used in the previous section) of the policies under two types
 235 of state-estimation noise at varying magnitudes. The first is zero-mean Gaussian noise, similar to the

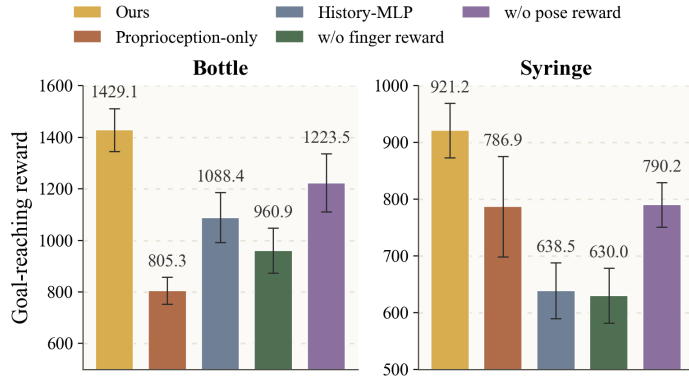


Figure 5: **Ablation study.** We compare our full policy against: (1) *proprioception-only*, which omits object state estimation; (2) *history MLP*, which replaces the LSTM with an MLP over 8 frames of input-output history; (3) *w/o finger reward*, trained without the finger-function contact reward $\rho^{\text{pinch}} \cdot \rho^{\text{grasp}}$; and (4) *w/o pose reward*, trained without the reference-pose reward ρ^{pose} .

domain randomization applied during training, which models random noise and occasional outlier observations. The second adds a constant mean shift to this baseline Gaussian noise throughout the episode; such systematic offsets are plausible in practice, arising, for example, from coordinate-frame misalignment between simulation and reality or from errors in camera-parameter estimation. We note that the per-axis noise standard deviation is set to 1 cm during training.

Fig. 6 reports the results for each setting with proprioception-only baselines. For both tasks, performance remains above 60% even under relatively strong noise (e.g., a 3 cm observation standard deviation), indicating considerable robustness to out-of-distribution observation noise. For the Syringe task, however, larger offset noise drives performance below the proprioception-only baseline. Given the performance of the proprioception-only policy, we conjecture that this robustness may stem from the policy’s maintenance of proprioceptive history.

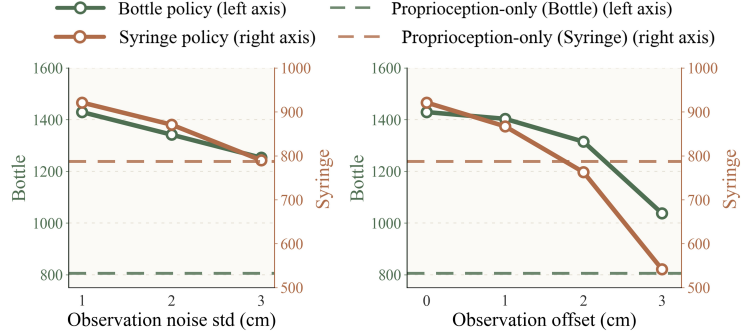


Figure 6: **Performance under different observation noise.** We report the goal-reaching reward under two noise types at varying magnitudes: zero-mean Gaussian noise (left), and the same Gaussian noise with a constant per-episode mean shift (right).

4.5 Real-World System Capabilities with a Single Camera

We conduct real-world experiments for the three tasks using only a depth camera. During the tasks, the fingers perform subtle in-hand reorientation of the objects to aid alignment and insertion, and the pinky finger is often used to increase insertion depth, as shown in Fig. 1.

Closed-loop control versus open-loop replay. To verify that closed-loop feedback is necessary, we run 20 consecutive trials comparing our closed-loop policy against open-loop replay of successful simulation trajectories. We record whether the *alignment* criterion (insertion depth over 1 cm) and the *assembly* criterion (final depth within 1 cm of the target) are met, as shown in Table 1. Open-loop replay achieves successful alignment only three times. Without feedback, the action sequences cannot compensate for sim-to-real dynamics, so early deviations accumulate, and the parts fail to align. The closed-loop policy succeeds far more often. A main failure case in the Bottle and Marker tasks is that an unfavorable initial cap pose prevents a stable pinch, leaving the cap tilted at an unrecoverable angle or dropped. In the Bottle and Syringe tasks, certain fingers can obstruct the insertion path, a state the policy has not learned to clear.

Table 1: **Closed-loop control versus open-loop replay.** Number of successful trials (out of 20) under the alignment and assembly criteria.

Method	Bottle		Syringe		Marker	
	Align	Assembly	Align	Assembly	Align	Assembly
Closed-loop (ours)	18	15	18	17	16	16
Open-loop replay	2	2	1	0	0	0

Robustness against Perturbation. We evaluate whether the policy has the ability to recover from unexpected deviations during the assembly process by artificially altering the state of one of the manipulated objects. Fig. 7 illustrates the policy’s recovery behaviors. In the Bottle task, the policy controls the index finger to correct an artificially induced misalignment of the bottle cap and realigns it without pinch. In the syringe task, we resist the hand’s grasping force to pull out the plunger, and the hand quickly restore it to the fully assembled state. These experiments validate that the policies have some robustness against unexpected state deviations.

Assembly under Various Hand Tilt. In Fig. 8, we present physical experiments demonstrating assembly with a single policy trained across various wrist-tilt angles, where the wrist pose is

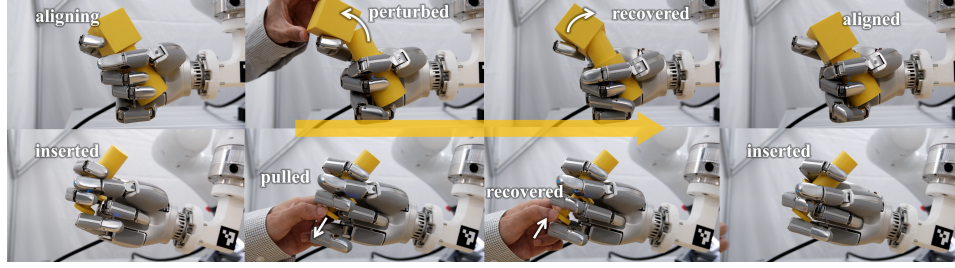


Figure 7: Policy behaviors under unexpected perturbation.

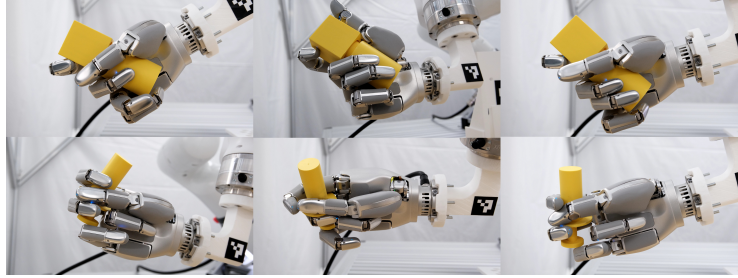


Figure 8: In-hand assembly under different hand tilts.

288 additionally provided as a policy observation. The policy succeeds across a range of tilt angles,
 289 showing that a single policy can adapt its finger coordination to the changing relative direction of
 290 gravity. However, this experiment reveals a limitation of our method: current rigid-body simula-
 291 tors [51] cannot model planar (patch) contacts, so a pinch grasp degenerates into two contact points
 292 and becomes unstable in simulation [52]. This makes it hard for the policy to reliably pinch-control
 293 $\mathcal{O}_h$ at certain tilt angles, e.g., when the object’s long axis starts near horizontal. Soft finger pads
 294 alleviate the instability in the real world, but introduce a sim-to-real gap. Consequently, the policy
 295 cannot learn decisive pinch behaviors in these configurations during simulation, leading to failures
 296 in real-world deployment.

297 **5 Limitations and Future Work**

298 A first limitation is that we study only a fragment of the in-hand manipulation workflow, rather than
 299 the full pipeline from grasping multiple objects off the table and adjusting their orientation for the
 300 subsequent assembly. The current isolation of in-hand manipulation requires artificial supporting
 301 force in some experiments until the policies engage, which may introduce factors that are difficult
 302 to model or quantify. However, a full workflow becomes long-horizon and involves skills, including
 303 multi-object grasping [19, 20] and reorienting objects from their initial placement to the desired in-
 304 hand pose [53], which remain open challenges. Second, we only currently include three tasks. In
 305 the future, we aim to implement more general part geometries and assembly types, such as threading
 306 and friction-fit, in the current suite to expand this work into a systematic benchmark for hardware
 307 and algorithms of in-hand manipulation.

308 **6 Conclusion**

309 We introduced in-hand assembly, where a single dexterous hand must both stabilize and assem-
 310 ble objects without external fixtures or a second arm. With its requirement for intricate contact
 311 transition and manipulation precision, this kind of task can serve as a benchmark for machine dex-
 312 terity. We propose a principled RL solution with lightweight reward design, which distinguishes
 313 the roles between fingers and differentiates this RL task from other in-hand manipulation tasks, and
 314 the real-world effectiveness across three different tasks is validated. These results suggest that an-
 315 thropomorphic dexterity is not merely a matter of many degrees of freedom, but can be realized
 316 through human-like functional finger specialization and coordinated contact use, moving robotic
 317 hands toward human-level multi-functional dexterous manipulation.

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